

SAFETY DATA SHEET (GHS)

Benzene | CAS: 71-43-2 | Formula: C₆H₆

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

DATA SOURCES CONSULTED

- ECHA Classification & Labelling Inventory — Benzene, CAS 71-43-2 (<https://echa.europa.eu/>)
- PubChem CID 241 — GHS Classification (Rev.9), <https://pubchem.ncbi.nlm.nih.gov/compound/241>
- ILO/NIOSH International Chemical Safety Card (ICSC) 0015 — Benzene
- ACGIH TLV/BEI — Benzene, 2024 Notice of Intended Change (TWA reduced to 0.02 ppm)
- NOAA CAMEO Chemicals — Benzene datasheet (<https://cameochemicals.noaa.gov/>)
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: Benzene

CAS Number: 71-43-2

Chemical formula: C₆H₆

Molecular weight: 78.11 g/mol

Synonyms: Benzol; Cyclohexatriene; Coal naphtha; Phene

Intended use: Feedstock/by-product stream component — Benzene/Cumene fractionation and recycle, alkylation unit feed circuits. PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification: Flam. Liq. 2 (H225); Skin Irrit. 2 (H315); Eye Irrit. 2 (H319); Germ Cell Mutag. 1B (H340); Carc. 1A (H350); STOT RE 1 (H372 — haematopoietic system/blood, via exposure route confirmed); Asp. Tox. 1 (H304); Aquatic Chronic 2 (H411)

Signal word: Danger

Hazard statements (H-codes): H225 (Highly flammable liquid and vapour); H304 (May be fatal if swallowed and enters airways); H315 (Causes skin irritation); H319 (Causes serious eye irritation); H340 (May cause genetic defects); H350 (May cause cancer — leukaemia); H372 (Causes damage to organs [blood/bone marrow, immune system] through prolonged or repeated exposure); H411 (Toxic to aquatic life with long lasting effects)

Precautionary statements (P-codes): P201, P202, P210, P233, P240, P241, P242, P243, P260, P264, P270, P273, P280, P301+P310, P303+P361+P353, P304+P340, P305+P351+P338, P308+P313, P331, P370+P378, P403+P233, P405, P501

GHS pictograms: GHS02 (Flammable), GHS08 (Health hazard), GHS07 (Irritant), GHS09 (Environmental hazard)

Other hazards: CRITICAL — Benzene is a confirmed human carcinogen (leukaemia) with no safe exposure threshold established for cancer endpoint; this is the single highest-priority chronic health hazard among the Train II process chemicals. ACGIH adopted a major TLV-TWA reduction (from the long-standing 0.5 ppm to 0.02 ppm) in its 2024 Notice of Intended Change cycle — confirm final adopted value and Thailand OEL alignment status before using for exposure assessments; this represents an order-of-magnitude tightening with significant implications for LEL/PPE/monitoring requirements at any benzene-containing stream sample points, vents, or seals. Vapours are heavier than air and can travel along the ground to ignition sources. Static-accumulating liquid.

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Benzene: CAS 71-43-2 — content per process stream specification (recycle/feed benzene streams typically >99% with trace toluene/cumene — confirm against PPCL stream compositions for benzene recycle and BC column overheads)

SECTION 4 — FIRST AID MEASURES

Inhalation: Remove to fresh air immediately. If breathing has stopped, provide artificial respiration; administer oxygen if available. Obtain immediate medical attention — acute high exposure can cause CNS depression, dizziness, narcosis, and

cardiac arrhythmia (sensitisation); chronic exposure causes haematological damage.

Skin contact: Remove contaminated clothing immediately. Wash skin thoroughly with soap and water. Obtain medical attention if irritation persists. Benzene is readily absorbed through intact skin — treat any significant dermal exposure as a systemic exposure event.

Eye contact: Flush immediately with copious water for at least 15 minutes, holding eyelids open. Obtain medical/ophthalmological attention.

Ingestion: Do NOT induce vomiting — high aspiration hazard (H304); aspiration into lungs can cause chemical pneumonitis and is potentially fatal. Obtain immediate medical attention. If vomiting occurs spontaneously, keep head below hips to prevent aspiration.

Notes to physician: Monitor for CNS depression/arrhythmia in acute exposure; for occupational exposure surveillance, consider baseline and periodic CBC (complete blood count) given haematotoxicity/leukaemogenicity. Urinary biomarkers (e.g. S-phenylmercapturic acid, t,t-muconic acid) may be used for biological monitoring per institutional protocol.

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Dry chemical, CO₂, alcohol-resistant foam, water spray/fog for cooling; do not use straight water streams which may spread fire

Unsuitable extinguishing media: High-volume straight water jets directly onto burning liquid (spreads fire due to low density vs. water)

Special hazards: Highly flammable liquid and vapour (Flash point -11°C). Vapours are heavier than air (vapour density ~2.7), can travel along the ground/floor to remote ignition sources and flash back. Closed containers may rupture/BLEVE if exposed to fire due to vapour pressure buildup. Significant static electricity accumulation hazard during transfer.

Special protective equipment for firefighters: Full bunker gear with positive-pressure SCBA; cool exposed containers/vessels with water spray from a safe distance; approach from upwind

Hazardous combustion products: Carbon monoxide, carbon dioxide, and various partial oxidation products (smoke is heavily sooting due to high carbon content)

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Eliminate all ignition sources (no smoking, open flames, non-rated electrical equipment). Evacuate non-essential personnel; use explosion-proof equipment. Wear appropriate respiratory protection — even low-level vapour exposure is a chronic carcinogen concern. Bond/ground all equipment during transfer to control static discharge.

Environmental precautions: Prevent entry to drains, sewers, watercourses and soil — toxic to aquatic life (H411) and a groundwater contamination concern (mobile, low water solubility forms separate phase).

Containment and cleanup: Contain spill with non-combustible absorbent (sand, vermiculite, commercial sorbent); do not use combustible materials. Collect in closed, labelled, explosion-proof-area containers for disposal. Ventilate area; monitor vapour concentration before re-entry.

Reference to other sections: See Section 7 (Handling/Storage), Section 8 (PPE incl. respiratory protection for carcinogen), Section 9 (flammability data)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Handle in closed systems; minimise opportunities for vapour release at sample points, pump seals, and relief devices given carcinogen classification. Use only explosion-proof/intrinsically safe electrical equipment and tools in benzene-handling areas. Bond and ground all transfer lines/containers to prevent static discharge. Avoid generating mists/vapours.

Storage conditions: Store in closed containers/tanks in a well-ventilated area, away from heat, sparks, open flames and other ignition sources. Tanks should have appropriate flame arrestors/conservation venting and be electrically bonded/grounded.

Temperature limits: Store below ambient where practical to limit vapour generation; keep well below flash point margin in enclosed spaces

Pressure: Atmospheric/low-pressure storage with conservation vent to vapour recovery/flare where required by environmental permit

Segregation requirements: Segregate from strong oxidisers, strong acids (especially nitric/sulfuric acid mixtures — nitration reaction risk), and incompatible reactive chemicals. Segregate from sources of ignition.

Special requirements: Given Carc. 1A / Mutag. 1B classification, minimise the number of personnel with potential exposure, implement a benzene exposure monitoring programme (personal sampling) at sample points, pump seals and tank vents, and ensure LEL monitoring is in place given the low flash point.

Specific end-use recommendations: Confirm benzene content of recycle/feed streams against the assumptions used in this SDS; any stream with appreciable benzene content should be treated under this SDS for exposure-control purposes even if benzene is a minor component.

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): ACGIH TLV-TWA: historically 0.5 ppm with STEL 2.5 ppm and Skin notation, A1 (confirmed human carcinogen); ACGIH's 2024 Notice of Intended Changes proposes/adopts a substantially lower TWA of 0.02 ppm — CONFIRM the final adopted ACGIH value and current Thailand statutory OEL (Ministry of Labour) before use, as this is a significant change with major implications for monitoring strategy and PPE selection at this facility. OSHA PEL: 1 ppm TWA / 5 ppm STEL (action level 0.5 ppm).

TWA: 0.02 ppm (ACGIH 2024 NIC, pending confirmation) vs. historical 0.5 ppm (ACGIH)/1 ppm (OSHA PEL) — FLAG FOR VERIFICATION

STEL: 2.5 ppm (ACGIH, historical value — confirm against 2024 NIC outcome)

Ceiling: Not established (OSHA Ceiling does not apply; some jurisdictions set ceiling/excursion limits)

Skin notation: Yes — significant dermal absorption contributes to systemic dose

IARC/NTP/OSHA carcinogen status: IARC Group 1 (carcinogenic to humans); NTP — known human carcinogen; OSHA regulates benzene under a substance-specific standard (29 CFR 1910.1028)

Engineering controls: Closed systems, vapour recovery on tanks, local exhaust ventilation at sample points and pump seals, continuous/periodic LEL and benzene-specific vapour monitoring

PPE — Respiratory: Air-purifying respirator with organic vapour cartridge for short-duration tasks below assigned protection factor limits; supplied-air respirator or SCBA for tank entry, high-concentration tasks, or emergency response — select based on current exposure monitoring data and the revised (lower) OEL

PPE — Gloves: Chemical-resistant gloves with demonstrated breakthrough time for benzene (e.g. laminate/Viton-type materials); nitrile/butyl rubber may have limited breakthrough — verify against manufacturer permeation data

PPE — Eyes: Chemical splash goggles; face shield for tasks with splash potential

PPE — Body: Flame-resistant (FR), chemical-resistant coveralls; antistatic footwear in classified areas

Biological monitoring: Urinary S-phenylmercapturic acid (S-PMA) and/or t,t-muconic acid as biomarkers of exposure, per occupational health programme

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Colourless to light-yellow liquid

Odour: Characteristic aromatic/sweet, gasoline-like odour

Odour threshold: ~1.5-5 ppm (odour is not a reliable warning property at the revised low OEL of 0.02 ppm)

pH: Not applicable (organic liquid, immiscible with water)

Melting point / Freezing point: 5.5°C

Boiling point (at 1 atm): 80.1°C

Flash point: -11°C (closed cup) — experimental literature value

Evaporation rate: ~4.5-5 (n-butyl acetate = 1) — evaporates significantly faster than water

Flammability (solid/gas): Highly flammable liquid

Flammability limits in air: LEL 1.2 vol%; UEL 7.8 vol%

Vapour pressure (at 20°C): ~10 kPa (~75 mmHg); ~12.7 kPa at 25°C

Vapour density (air = 1): ~2.7 (heavier than air — accumulates at grade/low points)

Relative density (water = 1): 0.8765 (at 20°C) — less dense than water, floats and spreads on water surface

Solubility (water): ~1.8 g/L at 25°C (slightly soluble)

Partition coefficient (log Kow): ~2.13

Auto-ignition temperature: ~498°C

Decomposition temperature: Not applicable under normal use; thermal decomposition at very high temperatures yields soot/PAH-type products and CO/CO₂ under combustion conditions

Viscosity: ~0.6 mPa·s at 25°C (low viscosity, mobile liquid)

Explosive properties: Not classified as explosive (vapour/air mixtures within LEL-UEL are flammable, not classified as an explosive substance per se)

Oxidising properties: Not an oxidiser

Molecular weight: 78.11 g/mol

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: Stable under normal conditions; reactive with strong oxidisers and in nitration/sulfonation reactions with strong acid mixtures

Chemical stability: Stable under recommended storage and handling conditions

Possibility of hazardous reactions: Can react vigorously/exothermically with strong oxidising agents (e.g. concentrated nitric acid, perchlorates, peroxides); mixtures with concentrated sulfuric/nitric acid can undergo exothermic nitration reactions; vapour/air mixtures within flammable range present explosion hazard in confined spaces if an ignition source is present

Conditions to avoid: Heat, sparks, open flames, static discharge, and other ignition sources; contact with strong oxidisers and strong acids

Incompatible materials: Strong oxidisers (nitric acid, perchlorates, peroxides), strong acids (sulfonation/nitration risk), some plastics/elastomers (swelling/permeation)

Hazardous decomposition products: None under normal storage; combustion produces CO, CO₂ and dense smoke/soot (PAHs)

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Inhalation (primary route of concern), dermal absorption (significant), ingestion (aspiration hazard)

Acute toxicity — Oral LD₅₀: ~930 mg/kg, rat

Acute toxicity — Dermal LD₅₀: >8,260 mg/kg, rabbit (low acute dermal lethality, but dermal absorption contributes to systemic/chronic dose)

Acute toxicity — Inhalation LC₅₀: ~10,000 ppm (4 h), rat (acute lethality threshold is far above the chronic OEL of concern)

Skin corrosion/irritation: Causes skin irritation (H315); defatting effects on prolonged/repeated contact

Serious eye damage/irritation: Causes serious eye irritation (H319)

Respiratory/skin sensitisation: Not classified as a sensitiser

Germ cell mutagenicity: May cause genetic defects (H340, Category 1B) — based on evidence of chromosomal damage (e.g. increased micronuclei) in exposed workers

Carcinogenicity: May cause cancer (H350, Category 1A) — established human leukaemogen (acute myeloid leukaemia); IARC Group 1. THIS IS THE MOST SIGNIFICANT CHRONIC HEALTH HAZARD AMONG TRAIN II PROCESS CHEMICALS — no established safe threshold for the carcinogenic endpoint.

Reproductive toxicity: Not formally classified under GHS for reproductive toxicity, though haematotoxic effects (H372) may have downstream relevance — confirm current ECHA classification status

STOT (single exposure): High concentrations cause CNS depression — headache, dizziness, drowsiness, narcosis; very high concentrations can cause cardiac sensitisation/arrhythmia and death

STOT (repeated exposure): Causes damage to blood-forming organs/bone marrow (H372) through prolonged or repeated exposure — aplastic anaemia, pancytopenia, and leukaemia are the principal chronic concerns

Aspiration hazard: May be fatal if swallowed and enters airways (H304) — chemical pneumonitis risk

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC₅₀/EC₅₀): Fish LC₅₀ (96h) ~5-29 mg/L (species dependent); Daphnia EC₅₀ (48h) ~8-25 mg/L — Toxic to aquatic life with long lasting effects (H411)

Persistence and degradability: Biodegrades under aerobic conditions but can persist in groundwater/anaerobic environments; moderately persistent in subsurface due to low solubility/sorption

Bioaccumulative potential: Low to moderate (log K_{ow} ~2.13) — not expected to bioaccumulate significantly

Mobility in soil: Moderate-to-high mobility in soil/groundwater; forms separate non-aqueous phase liquid (NAPL) and can be a persistent groundwater contamination source if released

Other adverse effects: Significant groundwater contamination potential; regulated drinking-water contaminant in most jurisdictions due to carcinogenicity

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Dispose of via licensed hazardous waste incineration or other approved destruction method in accordance with Thailand Hazardous Substance Act B.E. 2535 and site waste management procedures; do not discharge to drains/sewers

Contaminated packaging: Drain thoroughly; dispose of as hazardous waste or return for reconditioning per supplier program

Regulatory reference: Thailand Hazardous Substance Act B.E. 2535; Basel Convention considerations for any cross-border waste movement

SECTION 14 — TRANSPORT INFORMATION

UN Number: UN 1114

Proper shipping name: BENZENE

Transport hazard class(es): Class 3 (Flammable liquid)

Packing group: II

Environmental hazards (marine pollutant): Yes — marine pollutant (Aquatic Chronic 2)

Special precautions for user: Ensure adequate bonding/grounding during loading/unloading; transport in vapour-tight, properly vented tanks; carcinogen labelling/documentation requirements may apply per national regulations

Bulk transport (MARPOL/IBC): Classify per IBC Code if shipped in bulk by sea

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below. NOTE: OSHA regulates benzene under a dedicated substance-specific standard (29 CFR 1910.1028) with its own PEL, monitoring, and medical surveillance requirements distinct from the general Z-1 table.

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0

Key changes from previous version: Initial issue — compiled for HAZOP PSI (CAS 71-43-2). FLAG: ACGIH 2024 Notice of Intended Changes proposes reducing the benzene TLV-TWA from 0.5 ppm to 0.02 ppm — verify final adopted value and Thailand statutory OEL before use in exposure assessments; this is an order-of-magnitude change with major monitoring/PPE implications.

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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